



Stochastic Simulations in Dynamic Soil–Structure Interaction

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Abstract. This paper deals with several numerical techniques that account for random excitations and random material parameters occurring in earthquake engineering. It focuses in sequence on the influence on the structural response of the variability of the incident field using filtering theory, on the soil variability by coupling Stochastic Finite Elements, integral operators and Monte-Carlo simulation, and finally on the influence on the site response of a random building distribution using periodic simulations. In each case, practical examples are given to help illustrate the numerical techniques and to underline the importance of randomness in earthquake engineering problems.

Sommario. Questo lavoro prende in considerazione diverse tecniche numeriche che portano in conto l'aleatorietà dell'eccitazione e dei parametri del materiale in problemi di ingegneria sismica. L'attenzione viene rivolta sequenzialmente all'influenza sulla risposta strutturale della variabilità del campo incidente utilizzando una teoria del filtro, alla variabilità del suolo analizzata tramite elementi finiti stocastici, operatori integrali e simulazioni Monte-Carlo, ed infine all'influenza sulla risposta del sito di una distribuzione aleatoria degli edifici analizzata tramite simulazioni periodiche. Per ciascun caso, vengono forniti esempi pratici volti ad illustrare le tecniche numeriche e ad evidenziare l'importanza delle aleatorietà nei problemi di ingegneria sismica.

Key words: Stochastic modelling, Periodic models, Scattering operators, Soil–structure interaction, Boundary element methods.

1. Introduction

Soil–structure interaction [11, 37] has received attention in recent years due to the significant role it plays on the seismic response of massive structures such as nuclear power plants [18], arch dams or even smaller buildings resting on soft deposits [25, 31]. Earthquakes are uncertain events not only regarding their occurrence and location (not addressed in this paper) but also regarding their characteristics. Indeed, recordings of strong motion arrays show a stochastic behaviour of the seismic field both in the time and space variables, even over short distances [38] and especially in urban areas.

As first shown by Aki and Chouet [1] the variability of the incident field is mainly due to heterogeneities in the soil itself. Classical results show three main phenomena affecting waves in such random media depending on the amplitude and the density of such fluctuations [26]. When small or weak fluctuations are concerned, waves propagate in the random medium as in an *effective* deterministic media. The phenomena can be accounted for using mean field theories, asymptotic expansions or auto-coherent models. Compared to wave in homogeneous media, the amplitude of the coherent wavefield is reduced due to energy diffusion towards an incoherent wave field. However the total amplitude is not necessarily reduced and significant fluctuations around the mean field can be observed. After some propagation distance, namely the mean free path, the wave field becomes almost incoherent and the *effective* approach fails. In this regime, *diffusion* models have to be considered [9]. For stronger fluctuations,

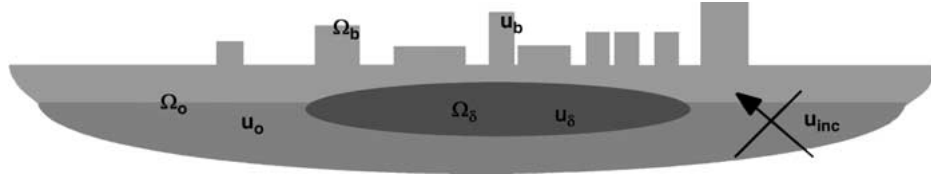


Figure 1. The generic model.

the medium generates strong back scattering of the wave field, giving rise to a partly coherent multiple scattering. The limiting case consists in the *localisation* phenomena classically occurring in a randomly layered media.

In practical situations it is almost impossible to choose between these regimes. Since contradictory conclusions on the amplitudes and thus on the safety of the structure are possible, more advanced studies have to be carried out. Fortunately, using present generation computers, random loadings and random parameters can be accounted for in the numerical simulation of soil–structure interaction. The objective of this paper is to develop a general numerical approach that can account for all these phenomena.

As a starting point of this study, a deterministic mathematical and numerical model for soil–structure interaction is given in Section 2, and is extended to periodic models in Section 3. The use of a stochastic incident field in this model is briefly described in Section 4 and the reader is referred to Clouteau and Aubry [13] for more details. Taking account of soil randomness in a bounded region of the numerical model is proposed in Section 5. The proposed method is based on a variational form of the Lippman Schwinger equation classically occurring in wave propagation in random media. It is solved numerically coupling finite elements and integral operators inside a bounded volume and performing Monte-Carlo simulations on a reduced numerical model.

This method does not apply when unbounded random fluctuations have to be accounted for. In order to study the influence on the seismic field of a large number of buildings randomly distributed inside a city, a third approach is proposed in Section 6. Assuming that the distribution of the buildings is a homogeneous random process along the free surface, it is shown that periodic models can be efficiently used. From a practical point of view, new results on localisation of the scattered field are given.

2. The Basic Deterministic Model

This section concentrates on a typical dynamic soil–structure interaction model assuming deterministic properties either for the physical parameters and for the loads. This model can be considered as a given realisation of the stochastic model that will be accounted for in subsequent sections. This model and the basic governing equations are given in Subsection 2.1 together with the underlying hypotheses. Subsections 2.2 and 2.3 are devoted to mathematical and numerical techniques to model such a physical problem using dual domain decomposition techniques and integral operators.

2.1. THE PHYSICAL MODEL

Let Ω_o be a stratified half-space with *effective* deterministic physical properties denoted by $\mathbf{p}_o = (\lambda_o, \mu_o, \rho_o)$ (Lamé elastic coefficients and the mass density) and a planar free surface

denoted by P_o . Inside Ω_o , a bounded region denoted by Ω_δ , has physical properties fluctuating around the *effective* properties $\mathbf{p}_o$. These fluctuations are denoted by $\mathbf{p}_\delta = (\lambda_\delta, \mu_\delta, \rho_\delta)$. Hence, the soil domain denoted by Ω_s , can be considered as the superposition of Ω_o and Ω_δ , its physical properties $\mathbf{p}_s = (\lambda_s, \mu_s, \rho_s)$ satisfying:

$$\begin{aligned}\mathbf{p}_s &= \mathbf{p}_o + \mathbf{p}_\delta \quad \text{inside } \Omega_\delta, \\ \mathbf{p}_s &= \mathbf{p}_o \quad \text{inside } \Omega_o \setminus \Omega_\delta.\end{aligned}\tag{1}$$

For the sake of simplicity it is assumed that $\partial\Omega_\delta \cap P_o = \emptyset$, where $\partial\Omega$ stands for the boundary of domain Ω . Buildings, generically denoted by Ω_b , are resting on Ω_s , $\Sigma \subset P_o$ denoting the planar interface between Ω_b and Ω_s . These structures have physical properties denoted by $\mathbf{p}_b$. The union of the sub-domains Ω_s and Ω_b is denoted by Ω .

In order to characterize the dynamic behaviour of Ω under seismic loadings, the following notation and hypotheses are assumed: $\mathbf{u}_\beta$ and $\mathbf{v}_\beta$ are, respectively the actual and virtual dynamic perturbation of displacement fields defined on Ω_β , for any subscript $\beta \in \{o, s, b, \delta\}$. Under the small displacement and small strain hypotheses, the dynamic perturbations of the stress tensor denoted by $\boldsymbol{\sigma}_\beta(\mathbf{v}_\beta)$ can be computed as a linear function of the strain tensor $\boldsymbol{\epsilon}(\mathbf{v}_\beta)$ using the Lamé coefficients λ_β and μ_β . The divergence of a second order tensor $\mathbf{A}$, denoted by $\text{Div } \mathbf{A}$, is the vector whose components in a cartesian frame are given by $\sum_{i=1,3} \partial_i A_{ij}$. Assuming the outer normal convention for the normal vector $\mathbf{n}$, the traction vector associated to any displacement field $\mathbf{u}_\beta$ is denoted by $\mathbf{t}_\beta(\mathbf{u}_\beta) = \boldsymbol{\sigma}_\beta(\mathbf{v}_\beta) \cdot \mathbf{n}$.

Next, the dynamic displacement field, $\mathbf{u}$ inside the overall domain Ω , has to satisfy the following Navier equation and the free surface boundary conditions on $\partial\Omega$ reading in the frequency domain:

$$\begin{aligned}\text{Div} \boldsymbol{\sigma}(\mathbf{u}) &= -\rho \omega^2 \mathbf{u} \quad \text{in } \Omega, \\ \mathbf{t}(\mathbf{u}) &= \mathbf{0} \quad \text{on } \partial\Omega.\end{aligned}\tag{2}$$

The seismic loading on Ω is due to an incident field denoted by $\mathbf{u}_o^{\text{inc}}$ in the Ω_o domain. Even in the stochastic case, it is assumed that this field satisfies the Navier equation in Ω_o and the free surface conditions on P_o :

$$\begin{aligned}\text{Div} \boldsymbol{\sigma}_o(\mathbf{u}_o^{\text{inc}}) &= -\rho_o \omega^2 \mathbf{u}_o^{\text{inc}} \quad \text{in } \Omega_o, \\ \mathbf{t}_o(\mathbf{u}_o^{\text{inc}}) &= \mathbf{0} \quad \text{on } P_o.\end{aligned}\tag{3}$$

Defining the *scattered field* $\mathbf{u}_d = \mathbf{u}_s - \mathbf{u}_o^{\text{inc}}$ as a new unknown field in the soil, the seismic loadings in equations (2) are generically defined inside Ω_δ as the following dynamic loads $\mathbf{f}^{\text{inc}}$:

$$\mathbf{f}_o^{\text{inc}} = \text{Div} \boldsymbol{\sigma}_\delta(\mathbf{u}_o^{\text{inc}}) + \rho_\delta \omega^2 \mathbf{u}_o^{\text{inc}} \quad \text{in } \Omega_\delta.\tag{4}$$

Hence, the *scattered field* $\mathbf{u}_d$ and the displacement field in the structure $\mathbf{u}_b$ have to satisfy:

$$\begin{aligned}\text{Div} \boldsymbol{\sigma}_s(\mathbf{u}_d) + \delta_{\Omega_\delta} \mathbf{f}_o^{\text{inc}} &= -\rho_s \omega^2 \mathbf{u}_d \quad \text{in } \Omega_s, \\ \text{Div} \boldsymbol{\sigma}_b(\mathbf{u}_b) &= -\rho_b \omega^2 \mathbf{u}_b \quad \text{in } \Omega_b, \\ \mathbf{t}_b(\mathbf{u}_b) + \mathbf{t}_s(\mathbf{u}_d) &= \mathbf{0} \quad \text{on } \Sigma, \\ \mathbf{u}_d - \mathbf{u}_b &= -\mathbf{u}_o^{\text{inc}} \quad \text{on } \Sigma,\end{aligned}\tag{5}$$

together with free surface conditions on the other boundaries. δ_{Ω_δ} stands for the characteristic function of Ω_δ inside Ω_s .

2.2. DOMAIN DECOMPOSITION AND INTEGRAL OPERATORS

Using classical Domain Decomposition techniques, the original problem (5) is transformed into a set of local problems coupled throughout interfaces that can be either a surface for the soil–structure interface Σ or the volume Ω_δ for the soil–fluctuation coupling. The two cases are dealt with in sequence.

2.2.1. Soil–structure coupling

As far as soil–structure interaction is concerned, classical dual domain decomposition approaches [2–3, 36] consist in taking the traction $\boldsymbol{\tau} = \mathbf{t}_b(\mathbf{u}_b) = -\mathbf{t}_s(\mathbf{u}_s)$ along the interface Σ as an auxiliary unknown field. The original problem (5) is then decomposed into two local problems with the interface traction $\boldsymbol{\tau}$ as an input parameter and are coupled through a kinematical condition on the Σ interface:

$$\begin{aligned} \mathbf{Div} \boldsymbol{\sigma}_s(\mathbf{u}_d) + \delta_{\Omega_\delta} \mathbf{f}_o^{\text{inc}} + \delta_\Sigma \boldsymbol{\tau} &= -\rho_s \omega^2 \mathbf{u}_d \quad \text{in } \Omega_s, \\ \mathbf{Div} \boldsymbol{\sigma}(\mathbf{u}_b) &= -\rho_b \omega^2 \mathbf{u}_b \quad \text{in } \Omega_b, \\ \mathbf{t}_b(\mathbf{u}_b) &= \boldsymbol{\tau} \quad \text{on } \Sigma, \\ \mathbf{u}_d - \mathbf{u}_b &= -\mathbf{u}_o^{\text{inc}} \quad \text{on } \Sigma, \end{aligned} \quad (6)$$

δ_Σ stands for the characteristic function of the Σ interface.

2.2.2. Soil–perturbation coupling

The procedure described for SSI is generalized here to couple the unbounded effective soil domain Ω_o with the bounded perturbed zone Ω_δ . Since these two domains are overlapping and even though it is a volume, Ω_δ plays the role of the interface. H. Bendhia [4] and J.L. Lions [29] have made similar developments using penalty techniques. $\mathbf{u}_\delta$ being the restriction of $\mathbf{u}_d$ on Ω_δ , the coupling force field ϕ between Ω_o and Ω_δ induced by the fluctuations of the mechanical properties is defined as follows:

$$\phi \stackrel{\text{def}}{=} \mathbf{Div} \boldsymbol{\sigma}_\delta(\mathbf{u}_\delta) + \rho_\delta \omega^2 \mathbf{u}_\delta + \mathbf{f}_o^{\text{inc}}. \quad (7)$$

Taking $\mathbf{u}_o$ as the restrictions of $\mathbf{u}_d$ on Ω_o and accounting for the momentum equation (first equation in system (6)) leads to the reciprocal form of ϕ as the opposite of the force applied by Ω_o on Ω_δ :

$$\delta_{\Omega_\delta} \phi = -(\mathbf{Div} \boldsymbol{\sigma}_o(\mathbf{u}_o) + \rho_o \omega^2 \mathbf{u}_o + \delta_\Sigma \boldsymbol{\tau}). \quad (8)$$

Provided with this new coupling field, fields $\mathbf{u}_o$, $\mathbf{u}_\delta$ and $\mathbf{u}_b$ have to satisfy the following equations on each sub-domain with $\boldsymbol{\tau}$ and ϕ as parameters:

$$\begin{aligned} \mathbf{Div} \boldsymbol{\sigma}_o(\mathbf{u}_o) + \delta_{\Omega_\delta} \phi + \delta_\Sigma \boldsymbol{\tau} &= -\rho_s \omega^2 \mathbf{u}_o \quad \text{in } \Omega_o, \\ \mathbf{Div} \boldsymbol{\sigma}_\delta(\mathbf{u}_\delta) - \phi + \mathbf{f}_o^{\text{inc}} &= -\rho_\delta \omega^2 \mathbf{u}_\delta \quad \text{in } \Omega_\delta, \\ \mathbf{t}_\delta(\mathbf{u}_\delta) &= \mathbf{0} \quad \text{on } \partial\Omega_\delta, \\ \mathbf{Div} \boldsymbol{\sigma}_b(\mathbf{u}_b) &= -\rho_b \omega^2 \mathbf{u}_b \quad \text{in } \Omega_b, \\ \mathbf{t}_b(\mathbf{u}_b) &= \boldsymbol{\tau} \quad \text{on } \Sigma. \end{aligned} \quad (9)$$

Moreover these fields have to satisfy the two coupling equations on Σ and Ω_δ , respectively:

$$\begin{aligned} \mathbf{u}_o - \mathbf{u}_b &= -\mathbf{u}_o^{\text{inc}} \quad \text{on } \Sigma, \\ \mathbf{u}_o - \mathbf{u}_\delta &= \mathbf{0} \quad \text{on } \Omega_\delta. \end{aligned} \quad (10)$$

These equations have to be solved for $\boldsymbol{\tau}$ and ϕ and this is first done by eliminating $\mathbf{u}_o$ using integral operators.

2.2.3. Integral operators

It is assumed that $\mathbf{U}^{Go}(\mathbf{x}, \mathbf{y})$, the fundamental solution of a layered half-space, exists and can be computed either analytically or numerically [12, 13] for any couple of points $(\mathbf{x}, \mathbf{y})$ in Ω_o . Hence, as long as Σ and Ω_δ are bounded and regular, the first equation of system (9) can be solved in terms of a single layer potential $\mathcal{U}_\Sigma^o$ and a Newton potential $\mathcal{U}_\delta^o$ as follows:

$$\begin{aligned}\mathbf{u}_o &= \mathcal{U}_\delta^o \phi + \mathcal{U}_\Sigma^o \boldsymbol{\tau}, \\ (\mathcal{U}_\delta^o \phi)(\mathbf{x}) &= \int_{\Omega_\delta} \mathbf{U}^{Go}(\mathbf{x}, \mathbf{y}) \phi(\mathbf{y}) dV_{\mathbf{y}}, \\ (\mathcal{U}_\Sigma^o \boldsymbol{\tau})(\mathbf{x}) &= \int_{\Sigma} \mathbf{U}^{Go}(\mathbf{x}, \mathbf{y}') \boldsymbol{\tau}(\mathbf{y}') dS_{\mathbf{y}'},\end{aligned}\tag{11}$$

Accounting for this decomposition, the coupling equations (10) now read:

$$\begin{aligned}\mathcal{U}_{\Sigma\Sigma}^o \boldsymbol{\tau} + \mathcal{U}_{\Sigma\delta}^o \phi - \mathbf{u}_b &= -\mathbf{u}_o^{\text{inc}} \quad \text{on } \Sigma, \\ \mathcal{U}_{\delta\Sigma}^o \boldsymbol{\tau} + \mathcal{U}_{\delta\delta}^o \phi - \mathbf{u}_\delta &= \mathbf{0} \quad \text{on } \Omega_\delta,\end{aligned}\tag{12}$$

where in the first equation $\mathcal{U}_{\Sigma\Sigma}^o$ and $\mathcal{U}_{\Sigma\delta}^o$ are the traces on Σ of the single layer potential $\mathcal{U}_\Sigma^o$ and the Newton potential $\mathcal{U}_\delta^o$, respectively. In the second equation $\mathcal{U}_{\delta\Sigma}^o$ and $\mathcal{U}_{\delta\delta}^o$ are the restrictions of these two potentials to Ω_δ . From standard integral operator theory [30], $\mathcal{U}_{\Sigma\Sigma}^o$ can be inverted in proper functional spaces which means that for any $\mathbf{u}_\Sigma$ defined on Σ , there exists a unique traction field $\boldsymbol{\tau}'$ on Σ such that $\mathbf{u}_\Sigma = \mathcal{U}_{\Sigma\Sigma}^o \boldsymbol{\tau}'$. Using the same arguments, it can be shown [34] that $\mathcal{U}_{\delta\delta}^o \phi$ can also be inverted and so, for any $\mathbf{v}_\delta$ on Ω_δ , there exists a unique force density ϕ' on Ω_δ such that $\mathbf{v}_\delta = \mathcal{U}_{\delta\delta}^o \phi'$ on Ω_δ .

Up to now the coupling procedures between, on the first hand, the soil and the structures and, on the second hand, the soil and the fluctuations have followed the same lines. This symmetry is lost in the following developments because Ω_b and Ω_δ have very different features. Indeed, Ω_b is usually ‘stiffer’ than Ω_o in the sense that $\mathbf{u}_b$ has to satisfy some kinematical conditions-rigid foundation for instance. On the contrary Ω_δ is often much ‘softer’ than Ω_o in the sense that the physical properties of Ω_δ are fluctuations that are not necessarily positive. Thus a physical problem on Ω_δ with properties $\mathbf{p}_\delta$ is not well-posed. As a consequence, these two cases will be addressed separately defining first a transmission and scattering operators on interface Σ and then a scattering operator on Ω_δ .

2.2.4. Transmission and scattering operators on the buildings

The first equation in (12) defines $\boldsymbol{\tau}$ as a linear function of $\mathbf{u}_b$. When accounted for in the last two equations of (9), $\mathbf{u}_b$ becomes a linear function of the total incident field on Σ , $\mathbf{u}_o^{\text{inc}} + \mathcal{U}_{\Sigma\delta}^o \phi$. This linear relationship defines the transmission operator $\mathcal{T}_\Sigma$ that formally reads:

$$\mathbf{u}_b = \mathcal{T}_\Sigma(\mathbf{u}_o^{\text{inc}} + \mathcal{U}_{\Sigma\delta}^o \phi).\tag{13}$$

Moreover, accounting for (12) in equation (11) leads to an expression of $\mathbf{u}_o$ as a linear function of $\mathbf{u}_o^{\text{inc}} + \mathcal{U}_{\Sigma\delta}^o \phi$. It formally defines the scattering operator $\mathcal{S}_\Sigma$ as follows:

$$\mathbf{u}_o = (\mathcal{I} + \mathcal{S}_\Sigma)(\mathbf{u}_o^{\text{inc}} + \mathcal{U}_{\Sigma\delta}^o \phi),\tag{14}$$

where $\mathcal{I}$ is the identity operator.

Formally, the scattering of the wave field on the buildings is now accounted for by $\mathcal{S}_\Sigma$ and so, $\mathbf{u}_o$, the total field in the soil, can be expressed as the sum of a linear operator $\mathcal{U}_\delta$ applying on ϕ and a new incident field denoted by $\mathbf{u}^{\text{inc}}$ satisfying coupling equations on the soil–structure interface:

$$\mathbf{u}_s = \mathbf{u}_o + \mathbf{u}_o^{\text{inc}} \stackrel{\text{def}}{=} \mathcal{U}_\delta \phi + \mathbf{u}^{\text{inc}}, \quad (15)$$

$$\mathcal{U}_\delta = (\mathcal{I} + \mathcal{S}_\Sigma) \mathcal{U}_{\Sigma\delta}^o, \quad \mathbf{u}^{\text{inc}} = (\mathcal{I} + \mathcal{S}_\Sigma) \mathbf{u}_o^{\text{inc}}. \quad (16)$$

Approximations for the operators $\mathcal{T}_\Sigma$, $\mathcal{S}_\Sigma$ and $\mathcal{U}_\delta$ in terms of integral operators are worked out in Subsection 2.3. In particular, $\mathbf{U}^G$, the kernel of $\mathcal{U}_\delta$ will be given.

2.2.5. Scattering operator on the fluctuation and Lippman-Schwinger equation

Once soil–structure interaction is accounted for by means of both operator $\mathcal{U}_\delta$ and the new incident field $\mathbf{u}^{\text{inc}}$, the interaction between the composite soil–structure domain and the heterogeneities Ω_δ is governed by the two remaining equations in (9). Then taking $\mathbf{u}_s = \mathbf{u}_\delta + \mathbf{u}_o^{\text{inc}}$ as the unknown field in the soil, these equations read:

$$\begin{aligned} \text{Div} \boldsymbol{\sigma}_\delta(\mathbf{u}_s) - \phi &= -\rho_\delta \omega^2 \mathbf{u}_s, \quad \text{in } \Omega_\delta, \\ \mathbf{t}_\delta(\mathbf{u}_s) &= \mathbf{0}, \quad \text{on } \partial\Omega_\delta, \\ \mathcal{U}_\delta \phi &= \mathbf{u}_s - \mathbf{u}^{\text{inc}}, \quad \text{on } \Omega_\delta. \end{aligned} \quad (17)$$

Classically eliminating ϕ in this equation leads to the Lippman-Schwinger equation for $\mathbf{u}_s$ [22]:

$$\mathbf{u}_s = \mathbf{u}^{\text{inc}} + \mathcal{S}_\delta \mathbf{u}. \quad (18)$$

When only density fluctuations are accounted for, the scattering operator $\mathcal{S}_\delta$ takes the following expression as a function of $\mathbf{U}^G$ the Green's tensor associated to $\mathcal{U}_\delta$:

$$(\mathcal{S}_\delta \mathbf{u})(\mathbf{x}) = -\omega^2 \int_{\Omega_\delta} \rho_\delta \mathbf{U}^G(\mathbf{x}, \mathbf{y}) \cdot \mathbf{u}(\mathbf{y}) \, dS_{\mathbf{y}}. \quad (19)$$

It is shown in Subsection 2.3 that this equation can be solved numerically in terms of ϕ rather than in terms of $\mathbf{u}$. Accounting for the last equation in (17), it reads:

$$\mathcal{U}_\delta \phi = \mathcal{S}_\delta (\mathcal{U}_\delta \phi - \mathbf{u}^{\text{inc}}). \quad (20)$$

This expression can be generalized to fluctuation of the elastic parameters using finite element discretization of Ω_δ and does not require the explicit derivatives of the kernel $\mathbf{U}^G$. Having a weakly singular numerical version of the Lippman-Schwinger equation is an important result when analysing random fluctuation of the soil parameters and it is shown in Section 5 how it can be used for stochastic simulations.

2.3. NUMERICAL APPROXIMATION

The numerical approximation in the proposed formulation is classically performed looking for the unknown fields $\boldsymbol{\tau}$, $\mathbf{u}_b$, ϕ and $\mathbf{u}_\delta$ in finite dimension spaces and writing kinematical conditions (12) and equilibrium equations (9) in a weak sense.

2.3.1. Soil–structure interaction

The traction field $\boldsymbol{\tau}$ on the interface Σ is approximated on a given basis $\boldsymbol{\Psi}(\mathbf{y}') = \{\boldsymbol{\psi}_n(\mathbf{y}')\}_{n=1, N_\Sigma}$, $\mathbf{q}_\Sigma$ being the vector of the coefficients of $\boldsymbol{\tau}$ on the basis: $\boldsymbol{\tau}(\mathbf{y}') = \boldsymbol{\Psi}(\mathbf{y}') \mathbf{q}_\Sigma$.

This vector is determined as a function of $\mathbf{u}_b$, ϕ and $\mathbf{u}_o^{\text{inc}}$ using a Galerkin procedure on the first coupling equation in (12):

$$\mathbf{U}_{\Sigma\Sigma}^o \mathbf{q}_\Sigma = \int_{\Sigma} \boldsymbol{\Psi}^T(\mathbf{y}') \cdot \{\mathbf{u}_b(\mathbf{y}') - (\mathbf{u}_o^{\text{inc}} + \mathcal{U}_{\Sigma\delta}^o \phi)(\mathbf{y}')\} d\mathbf{S}_{\mathbf{x}}, \quad (21)$$

$$\text{with } [\mathbf{U}_{\Sigma\Sigma}^o]_{nm} = \int_{\Sigma} \int_{\Sigma} \psi_n^T(\mathbf{y}') \mathbf{U}^{Go}(\mathbf{y}'', \mathbf{y}') \psi_m(\mathbf{y}'') dS_{\mathbf{y}'} dS_{\mathbf{y}''}. \quad (22)$$

At the same time, a variational formulation of the equilibrium equations on Ω_b is considered. Using a finite dimension basis $\mathbf{V}(\mathbf{x}_b) = \{\mathbf{v}_n(\mathbf{x}_b)\}_{n=1, N_b}$ to approximate the displacement field

$$\mathbf{u}_b(\mathbf{x}_b) = \mathbf{V}(\mathbf{x}_b) \mathbf{q}_b, \quad (23)$$

and taking virtual displacement fields in the same basis, the vector of the generalized degrees of freedom $\mathbf{q}_b$ has to satisfy the classical soil–structure interaction equation:

$$\underbrace{\{\mathbf{K}_s + \mathbf{K}_b - \omega^2 \mathbf{M}_b\}}_{\mathbf{Z}_{ob}} \mathbf{q}_b = \mathbf{f}(\phi). \quad (24)$$

In this equation, $\mathbf{K}_b$ and $\mathbf{M}_b$ are the classical stiffness and mass matrices of the structure for the basis $\mathbf{V}$ and are given by:

$$[\mathbf{K}_b]_{nm} = \int_{\Omega_b} \boldsymbol{\sigma}_b(\mathbf{v}_n) : \boldsymbol{\epsilon}(\mathbf{v}_m) dV, \quad [\mathbf{M}_b]_{nm} = \int_{\Omega_b} \rho_b \mathbf{v}_n \cdot \mathbf{v}_m dV. \quad (25)$$

$\mathbf{K}_s$ is the classical soil impedance and $\mathbf{f}$ is the classical seismic force given by:

$$\mathbf{K}_s = \mathbf{B}^T (\mathbf{U}_{\Sigma\Sigma}^o)^{-1} \mathbf{B}, \quad \mathbf{B} = \int_{\Sigma} \boldsymbol{\Psi}^T(\mathbf{y}') \cdot \mathbf{V}(\mathbf{y}') dS_{\mathbf{y}'}, \quad (26)$$

$$\mathbf{f}(\phi) = \mathbf{B}^T (\mathbf{U}_{\Sigma\Sigma}^o)^{-1} \mathbf{q}^{\text{inc}}, \quad \mathbf{q}^{\text{inc}} = \int_{\Sigma} \boldsymbol{\Psi}^T(\mathbf{y}') \cdot (\mathbf{u}_o^{\text{inc}} + \mathcal{U}_{\Sigma\delta}^o \phi)(\mathbf{y}') dS_{\mathbf{y}'}. \quad (27)$$

The approximate solution $\mathbf{u}_b$ is then given as a linear function of $\mathbf{u}_{\Sigma}^{\text{inc}}$ throughout the transmission integral operator $\mathcal{T}_{\Sigma}$ on Σ with kernel $\mathbf{T}_{\Sigma}(\mathbf{x}_b, \mathbf{y}')$:

$$\mathbf{u}_b(\mathbf{x}_b) = (\mathcal{T}_{\Sigma} \mathbf{u}_{\Sigma}^{\text{inc}})(\mathbf{x}_b) = \int_{\Sigma} \mathbf{T}_{\Sigma}(\mathbf{x}_b, \mathbf{y}') \cdot \mathbf{u}_{\Sigma}^{\text{inc}}(\mathbf{y}') dS_{\mathbf{y}'}, \quad (28)$$

where $\mathbf{T}_{\Sigma}(\mathbf{x}_b, \mathbf{y}')$ is given by:

$$\mathbf{T}_{\Sigma}(\mathbf{x}_b, \mathbf{y}') = \mathbf{U}_{\Sigma}^b(\mathbf{x}_b) \boldsymbol{\Psi}^T(\mathbf{y}'), \quad \mathbf{U}_{\Sigma}^b(\mathbf{x}_b) = \mathbf{V}(\mathbf{x}_b) \mathbf{Z}_{ob}^{-1} \mathbf{B}^T (\mathbf{U}_{\Sigma\Sigma}^o)^{-1}. \quad (29)$$

Operator $\mathcal{T}_{\Sigma}$ will play a central role when studying the effect of a random incident field on the response of the structure as described in Section 4.

2.3.2. Fundamental solution and incident field in the soil–structure system

It has been noticed that $\mathbf{u}_o$, the total field in the soil, can be expressed as the sum of a linear operator $\mathcal{U}_{\delta}$ applying on ϕ and a new incident field $\mathbf{u}^{\text{inc}}$ satisfying coupling equation on the soil–structure interface. Using equations (11), (24) and (27), the scattering operator $\mathcal{S}_{\Sigma}$ can

be written as an integral operator applying on any displacement field $\mathbf{u}_\Sigma$ defined on the soil–structure interface Σ :

$$(\mathcal{S}_\Sigma \mathbf{u}_\Sigma)(\mathbf{x}_s) = \left(\int_\Sigma \mathbf{U}^{Go}(\mathbf{x}_s, \mathbf{y}') \boldsymbol{\Psi}(\mathbf{y}') dS_{\mathbf{y}'} \right) \mathbf{S}_\Sigma \left(\int_\Sigma \boldsymbol{\Psi}(\mathbf{y}'') \mathbf{u}_\Sigma(\mathbf{y}'') dS_{\mathbf{y}''} \right)^T, \quad (30)$$

where the first integral accounts for the emission in Ω_o due to equivalent sources on Σ and the second integral is simply the projection of the displacement field $\mathbf{u}_\Sigma$ on the dual basis $\boldsymbol{\Psi}$. Finally matrix $\mathbf{S}_\Sigma$ is the approximate scattering matrix of the soil–structure system relating the equivalent forces on the foundation with the projection of an incident displacement field on Σ . This matrix reads:

$$\mathbf{S}_\Sigma = (\mathbf{U}_{\Sigma\Sigma}^o)^{-1} \{ \mathbf{B} \mathbf{Z}_{ob}^{-1} \mathbf{B}^T - \mathbf{U}_{\Sigma\Sigma}^o \} (\mathbf{U}_{\Sigma\Sigma}^o)^{-1}, \quad (31)$$

and it is linearly dependent on the difference between the compliance of the soil, $\mathbf{U}_{\Sigma\Sigma}^o$, and the compliance of the structure $\mathbf{B} \mathbf{Z}_{ob}^{-1} \mathbf{B}^T$.

As a consequence, $\mathcal{U}_\delta$ can be written as a Newton potential of the composite soil–structure domain and its kernel $\mathbf{U}^G(\mathbf{x}_s, \mathbf{y})$ can be splitted into a singular part $\mathbf{U}^{Go}(\mathbf{x}_s, \mathbf{y})$, the first Green's tensor of the layered half-space and a regular part $\mathbf{U}_r^{Go}(\mathbf{x}_s, \mathbf{y})$ scattered by the structure with:

$$\mathbf{U}^G(\mathbf{x}_s, \mathbf{y}) = \mathbf{U}^{Go}(\mathbf{x}_s, \mathbf{y}) + \mathbf{U}_r^{Go}(\mathbf{x}_s, \mathbf{y}), \quad (32)$$

$$\mathbf{U}_r^{Go}(\mathbf{x}_s, \mathbf{y}) = \left(\int_\Sigma \mathbf{U}^{Go}(\mathbf{x}_s, \mathbf{y}') \cdot \boldsymbol{\Psi}(\mathbf{y}') dS_{\mathbf{y}'} \right) \mathbf{S}_\Sigma \left(\int_\Sigma \mathbf{U}^{Go}(\mathbf{y}, \mathbf{y}'') \cdot \boldsymbol{\Psi}(\mathbf{y}'') dS_{\mathbf{y}''} \right)^T. \quad (33)$$

2.3.3. Mixed soil-heterogeneity coupling

The key point here is to notice that in contrast to classical domain decomposition techniques, the first equation in (17) cannot be solved taking ϕ as a prescribed load. However the second equation in (17) defines a unique relationship between $\mathbf{u}_\delta$ and ϕ since the integral operator $\mathcal{U}_\delta$ can be inverted.

Therefore, the only difficult remaining point is to find a numerical approximation of these equations using a different finite dimension basis $\boldsymbol{\Phi}_\delta = \{\boldsymbol{\Phi}_{kn}(\mathbf{x}_\delta)\}_{n=1, N_\delta, k=1, 3}$ and $\mathbf{W}_\delta = \{\mathbf{w}_{iI}(\mathbf{x}_\delta)\}_{I=1, M_\delta, i=1, 3}$ for ϕ and $\mathbf{u}_\delta$, respectively. Indeed, these bases have to satisfy the classical compatibility condition for mixed systems [8]. It has been shown in Savin [34] that piecewise constant shape functions for ϕ and FEM linear shape functions for $\mathbf{u}_\delta$ satisfy this condition. Thus writing the first equation in a variational form, for any of the virtual field $\mathbf{v}'_\delta = \mathcal{U}_\delta \phi'$, the following final linear system is obtained for $\mathbf{q}_\delta$, the vector of coefficients of ϕ on basis $\boldsymbol{\Phi}_\delta$:

$$\mathbf{U} \mathbf{q}_\delta + \mathbf{S}_\delta (\mathbf{U}_\delta \mathbf{q}_\delta + \mathbf{u}_\delta^{\text{inc}}) = \mathbf{0}, \quad (34)$$

$$\mathbf{S}_\delta = \mathbf{U}_\delta^T (\mathbf{K}_\delta - \omega^2 \mathbf{M}_\delta). \quad (35)$$

Matrices $\mathbf{K}_\delta$ and $\mathbf{M}_\delta$ are the classical FE matrices of Ω_δ with properties $\mathbf{p}_\delta$, whereas matrices $\mathbf{U}$, $\mathbf{U}_\delta$ are two approximations of the Newton potential $\mathcal{U}_{\delta\delta}$ given by:

$$[\mathbf{U}]_{nkml} = \int_{C_n} \int_{C_m} U_{kl}^G(\mathbf{x}, \mathbf{y}) dV_{\mathbf{y}} dV_{\mathbf{x}}, \quad (36)$$

$$[U_\delta]_{liml} = \int_{C_m} U_{il}^G(\mathbf{x}_l, \mathbf{y}) dV_{\mathbf{y}}, \quad [\mathbf{u}_\delta^{\text{inc}}]_{lk} = \mathbf{u}^{\text{inc}}(\mathbf{x}_l) \cdot \mathbf{e}_i, \quad (37)$$

where $\mathbf{x}_l$ stands for the coordinates of the FEM nodes. Equation (34) is thus the approximate, on finite dimension spaces of the classical Lippman-Schwinger equation given in (20).

Once $\mathbf{q}_\delta$ is known solving the linear system (34) the total displacement field in the buildings is given by equation (13). Accounting for the approximation of ϕ on basis Φ_δ finally gives:

$$\begin{aligned} \mathbf{u}_b(\mathbf{x}_b) &= \mathbf{U}_\Sigma^b(\mathbf{x}_b) \mathbf{q}_o^{\text{inc}} + \mathbf{U}_\delta^b \mathbf{q}_\delta, \quad \mathbf{q}_o^{\text{inc}} = \int_\Sigma \Psi^T(\mathbf{y}) \cdot \mathbf{u}_o^{\text{inc}}(\mathbf{y}) dS_{\mathbf{y}}, \\ \mathbf{U}_\delta^b(\mathbf{x}_b) &= \mathbf{U}_\Sigma^b(\mathbf{x}_b) \mathbf{U}_{\Sigma\delta}, \quad [\mathbf{U}_{\Sigma\delta}]_{nkn'} = \int_\Sigma \int_{\Omega_\delta} \Psi_n^T(\mathbf{x}) \mathbf{U}^{Go}(\mathbf{x}, \mathbf{y}) \Phi_{kn'}(\mathbf{y}) dV_{\mathbf{y}} dS_{\mathbf{x}}. \end{aligned} \quad (38)$$

2.3.4. Limitations and extensions

Although quite general, the proposed numerical procedures are restricted to bounded ‘interfaces’. Hence, they are applicable neither to a large collection of buildings resting on the soil nor to unbounded heterogeneities in the soil. Periodic models proposed in the next section are an alternative to account for such cases. It has to be noticed that such periodic models will lead to auxiliary problems with bounded interface and thus, once the periodic decomposition is applied, the aforementioned procedures are applicable.

On the second hand, only deterministic incident fields and physical properties have been accounted for so far. The case of stochastic incident fields will be accounted for in Section 4 using the linear filtering theory whereas random properties of the soil in a bounded region, namely Ω_δ , are addressed in Section 5.

3. SSI for a Periodic Bidimensional Network

Based on Floquet’s pioneering work [21] on differential equations with periodic coefficients, this section shows that a dynamic SSI analysis on a periodic two dimensional network of buildings can be restricted to the analysis of a set of independent problems defined on a reference cell. Some mathematical details on these aspects can be found in Bensoussan *et al.*, Chen and Friedman, Sanchez-Hubert and Turbe, [5, 10, 33]. In the context of SSI it is described by Clouteau *et al.* for 1D periodicity and extended to 2D periodicity in Subsection 3.1. One can also refer to Bossavit [7] since the methodology is quite similar to the one used for symmetry conditions.

The key point in these developments consists in solving the boundary value problems on a reference cell using modified Finite and Boundary Elements techniques. As far as BEM is concerned, the only modification consists in using Green’s functions that satisfy periodicity conditions. It is shown in Subsection 3.2 that such fundamental solutions can be easily and efficiently built using the Floquet decomposition either for a homogeneous or a stratified visco-elastic half-space.

3.1. 2D GENERALIZED PERIODICITY

Let us assume here that the physical model under consideration and denoted by Ω in the previous section, is periodic along the two horizontal directions $\mathbf{e}_1$ and $\mathbf{e}_2$ with a vector period $\mathbf{l} = (l_1, l_2)^T$. With such a hypothesis, Ω can be decomposed into an infinite set of equivalent cells indexed by $\mathbf{n} = (n_1, n_2)^T \in \mathbb{Z}^2$ and the following convention is used: $\mathbf{ln} = \mathbf{e}_1 l_1 n_1 + \mathbf{e}_2 l_2 n_2$.

Using the Floquet transform as proposed in subsection 3.1.1, the analysis is restricted to the generic cell denoted by $\tilde{\Omega} = \{\mathbf{x} \in \Omega / 0 < \mathbf{x} \cdot \mathbf{e}_p < l_p, p = 1, 2\}$. In this cell, the restriction of the soil domain, the structure domain and the interface are, respectively denoted by $\tilde{\Omega}_s = \Omega_s \cap \tilde{\Omega}$, $\tilde{\Omega}_b = \Omega_b \cap \tilde{\Omega}$, $\tilde{\Sigma} = \Sigma \cap \tilde{\Omega}$. Boundaries of these subdomains consist not only in the restrictions of their original boundaries but also those of the new boundaries of the cell. For example $\partial\tilde{\Omega}_s$, the boundary of $\tilde{\Omega}_s$ reads:

$$\begin{aligned} \partial\tilde{\Omega} &= \tilde{P}_o \cup \Sigma_{o1} \cup \Sigma_{l1} \cup \Sigma_{o2} \cup \Sigma_{l2}, \\ \Sigma_{op} &= \{\mathbf{x} \in \Omega_s / \mathbf{x} \cdot \mathbf{e}_p = 0\}, \quad \Sigma_{lp} = \{\mathbf{x} \in \Omega_s / \mathbf{x} \cdot \mathbf{e}_p = l_p\}, \end{aligned} \quad (39)$$

where $\tilde{P}_o$ is the restriction of the planar free-surface P_o on the generic cell, and where Σ_{op} and Σ_{lp} are additional infinite boundaries on which periodic conditions will hold.

3.1.1. Floquet transform

On the reference cell, only loads satisfying some periodicity conditions can be applied. Hence, the incident field has to be decomposed into a set of pseudo-periodic fields $\tilde{\mathbf{u}}_o^{\text{inc}}(\tilde{\mathbf{x}}, \boldsymbol{\kappa})$ using the so-called Floquet transform:

$$\tilde{\mathbf{u}}_o^{\text{inc}}(\tilde{\mathbf{x}}, \boldsymbol{\kappa}) = \sum_{\mathbf{n} \in \mathbb{Z}^2} \mathbf{u}^{\text{inc}}(\tilde{\mathbf{x}} + \mathbf{l}\mathbf{n}) e^{i\boldsymbol{\kappa} \cdot \mathbf{l}\mathbf{n}}, \quad (40)$$

$\boldsymbol{\kappa}$ being a two component vector belonging to $[-\pi/l_1, \pi/l_1] \times [-\pi/l_2, \pi/l_2]$. The equivalent forces $\tilde{\mathbf{f}}_o^{\text{inc}}$ due to soil heterogeneities (see equation (4)) can also be decomposed into pseudo-periodic forces $\tilde{\mathbf{f}}_o^{\text{inc}}$ using the same transform or simply using definition (4) replacing $\mathbf{u}_o^{\text{inc}}$ by $\tilde{\mathbf{u}}_o^{\text{inc}}$. When dealing with incident plane waves characterized by their horizontal wavenumber $\mathbf{k}^{\text{inc}}$ the Floquet transform reduces to:

$$\begin{aligned} \tilde{\mathbf{u}}_o^{\text{inc}}(\tilde{\mathbf{x}}, \boldsymbol{\kappa}) &= \delta(\boldsymbol{\kappa} - \boldsymbol{\kappa}^{\text{inc}}) \hat{\mathbf{u}}_o^{\text{inc}}(\mathbf{k}^{\text{inc}}) e^{-i\boldsymbol{\kappa}^{\text{inc}} \cdot \tilde{\mathbf{x}}}, \\ \kappa_p^{\text{inc}} &= -k_p^{\text{inc}} \mod [2\pi/l_p], \quad p = 1, 2, \end{aligned} \quad (41)$$

and as a consequence, only one value of $\boldsymbol{\kappa}$ has to be accounted for.

3.1.2. Pseudo-periodic solutions

Applying the same transform on equation (5) leads to the following generic equations for $(\tilde{\mathbf{u}}_d(\tilde{\mathbf{x}}, \boldsymbol{\kappa}), \tilde{\mathbf{u}}_b(\tilde{\mathbf{x}}, \boldsymbol{\kappa}))$ defined on $\Omega \times \tilde{\Omega}_b$ and for any $\boldsymbol{\kappa}$:

$$\begin{aligned} \text{Div} \boldsymbol{\sigma}_s(\tilde{\mathbf{u}}_d) + \tilde{\mathbf{f}}_o^{\text{inc}} &= -\rho_s \omega^2 \tilde{\mathbf{u}}_d, \quad \text{in } \tilde{\Omega}_s, \\ \text{Div} \boldsymbol{\sigma}_b(\tilde{\mathbf{u}}_b) &= -\rho_b \omega^2 \tilde{\mathbf{u}}_b, \quad \text{in } \tilde{\Omega}_b \\ \mathbf{t}_s(\tilde{\mathbf{u}}_d) + \mathbf{t}_l(\tilde{\mathbf{u}}_b) &= \mathbf{0}, \quad \text{on } \tilde{\Sigma}, \\ \tilde{\mathbf{u}}_d - \tilde{\mathbf{u}}_b &= \tilde{\mathbf{u}}_o^{\text{inc}}, \quad \text{on } \tilde{\Sigma}, \\ e^{i\boldsymbol{\kappa} \cdot \mathbf{l}} \tilde{\mathbf{u}}_\alpha(\tilde{\mathbf{x}} + \mathbf{l}_i \mathbf{e}_i) &= \tilde{\mathbf{u}}_\alpha(\tilde{\mathbf{u}}), \quad \text{for } \tilde{\mathbf{x}} \in \Sigma_{oi}, \quad \alpha \in \{d, b\} \\ &\quad i = 1, 2. \end{aligned} \quad (42)$$

3.1.3. Inverse Floquet transform

The solution $\mathbf{u}_\alpha(\mathbf{x})$, $\alpha = s, b$, is recovered using the following inverse Floquet formula:

$$\mathbf{u}_\alpha(\mathbf{x} = \tilde{\mathbf{x}} + \mathbf{l}\mathbf{n}) = \frac{l_1 l_2}{4\pi^2} \int_{-\pi/l_1}^{\pi/l_1} \int_{-\pi/l_2}^{\pi/l_2} \tilde{\mathbf{u}}_\alpha(\tilde{\mathbf{x}}, \boldsymbol{\kappa}) e^{-i\boldsymbol{\kappa} \cdot \mathbf{l}\mathbf{n}} d\boldsymbol{\kappa}. \quad (43)$$

It has to be noticed that no integration is needed to recover this solution when plane incident waves are accounted for since only one value of κ is to be considered.

3.1.4. Numerical solution

The numerical procedure to solve problem (42) on the reference cell follows exactly the one given in Section 2.2. Indeed it may be noticed that $\tilde{\Sigma}$ and $\tilde{\Omega}_\delta$ are bounded and thus fields on these surfaces can be approximated using a finite number of basis functions. However since the additional boundaries Σ_{op} and Σ_{Lp} are unbounded, some developments have to be carried out. The key idea to get rid of these boundaries is to build basis and trial fields satisfying *a priori* the generalised periodicity conditions (last equation in system (42)). When dealing with bounded domains such as $\tilde{\Omega}_b$ and $\tilde{\Omega}_\delta$, being accounted for using standard Finite Elements, the periodicity condition can be accommodated using the same mesh on two opposite sides of the cell as proposed by Clouteau *et al.* [15]. For an unbounded domain such as Ω_s , periodic integral equations and boundary elements are proposed in Section 3.2. Then as far as periodicity constraints are satisfied for both trial and basis functions, equations (24) and (34) hold on the periodic cell. Scattering and transmission operators can also be defined and approximated.

3.2. PERIODIC BOUNDARY ELEMENTS

The aim of this subsection is to use boundary integral equation [17] and boundary elements [6] to compute approximate solutions of boundary value problems with periodic boundary conditions that is finding $\tilde{\mathbf{u}}$ on Ω_o satisfying:

$$\begin{aligned} \text{Div} \boldsymbol{\sigma}(\tilde{\mathbf{u}}) + \tilde{\mathbf{f}} &= -\rho\omega^2 \tilde{\mathbf{u}}, \quad \text{in } \tilde{\Omega}_s, \\ \tilde{\mathbf{u}} &= \tilde{\mathbf{u}}_o \quad \text{on } \tilde{\Sigma}, \\ e^{i\kappa \mathbf{l}} \tilde{\mathbf{u}}(\mathbf{x} + \mathbf{l}) &= \tilde{\mathbf{u}}(\mathbf{x}) \quad \text{for } \mathbf{x} \in \Sigma_{op}, p = 1, 2. \end{aligned} \quad (44)$$

Since boundaries carrying periodicity conditions can be unbounded, the use of a periodic fundamental solution $\tilde{\mathbf{U}}^G$ and associated integral operator $\tilde{\mathcal{U}}$ instead of the usual Green's function $\tilde{\mathbf{U}}^G$ have been proposed in Clouteau *et al.* [14]. $\tilde{\mathbf{U}}^G$ is defined as the Floquet transform of $\tilde{\mathbf{U}}^G$ with respect to the source argument $\mathbf{y}$:

$$\tilde{\mathbf{U}}^G(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}; \kappa) = \sum_{\mathbf{n} \in \mathbb{Z}^2} e^{i\kappa \cdot \mathbf{ln}} \mathbf{U}^G(\tilde{\mathbf{x}}, \tilde{\mathbf{y}} + \mathbf{ln}). \quad (45)$$

Integral operators $\tilde{\mathcal{U}}_\Sigma$ and $\tilde{\mathcal{U}}_\delta$ corresponding to the ones defined in equation (11) are then given by mean of this new kernel as follows:

$$\tilde{\mathcal{U}}_\Sigma(\tilde{\boldsymbol{\tau}})(\tilde{\mathbf{y}}) = \int_{\tilde{\Sigma}} \tilde{\mathbf{U}}^G(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \tilde{\boldsymbol{\tau}}(\tilde{\mathbf{x}}) dS(\tilde{\mathbf{x}}), \quad \tilde{\mathcal{U}}_\delta(\tilde{\boldsymbol{\phi}})(\tilde{\mathbf{y}}) = \int_{\tilde{\Omega}_\delta} \tilde{\mathbf{U}}^G(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \tilde{\boldsymbol{\phi}}(\tilde{\mathbf{x}}) dV(\tilde{\mathbf{x}}).$$

From these definitions, it can be easily noticed that $\tilde{\mathbf{U}}^G(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$, $\tilde{\mathcal{U}}_\Sigma(\tilde{\boldsymbol{\tau}})$ and $\tilde{\mathcal{U}}_\delta(\tilde{\boldsymbol{\phi}})$ are pseudo-periodic functions with respect to $\tilde{\mathbf{y}}$ with a wavenumber equal to κ . Moreover, locally $\tilde{\mathbf{U}}^G$ has the same singularities as $\mathbf{U}^G$ and thus no additional regularization procedure is required for the numerical integration.

The numerical approximation of integral operators $\tilde{\mathcal{U}}_\Sigma$ and $\tilde{\mathcal{U}}_\delta$ follows exactly the same lines as in Section 2. In particular matrix $\mathbf{U}_{\Sigma\Sigma}^o$ in equation (22) and matrices $\mathbf{U}$ and $\mathbf{U}_\delta$ in equation (34) can be computed using the new kernel $\tilde{\mathbf{U}}^G(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$. As an example:

$$[\tilde{\mathbf{U}}_{\Sigma\Sigma}^o]_{nm} = \int_{\tilde{\Sigma}} \int_{\tilde{\Sigma}} \boldsymbol{\psi}_n^T(\tilde{\mathbf{x}}) \tilde{\mathbf{U}}^G(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \boldsymbol{\psi}_m(\tilde{\mathbf{y}}) dS_{\tilde{\mathbf{y}}} dS_{\tilde{\mathbf{x}}}. \quad (46)$$

As long as the series (45), without the term $n=0$, converges uniformly for any $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$, the sum in (45) and the integral in (46) can be exchanged to get the following expression for $\tilde{\mathbf{U}}_{\Sigma\Sigma}^o$:

$$\tilde{\mathbf{U}}_{\Sigma\Sigma}^o = \sum_{n=-\infty}^{+\infty} e^{in\kappa L} \tilde{\mathbf{U}}_{\Sigma\Sigma}^{on}, \quad \tilde{\mathbf{U}}_{\Sigma\Sigma}^{on} = \int_{\tilde{\Sigma}} \int_{\tilde{\Sigma}} \boldsymbol{\psi}_n^T(\tilde{\mathbf{x}}) \tilde{\mathbf{U}}^G(\tilde{\mathbf{x}}, \tilde{\mathbf{y}} + n\mathbf{l}) \boldsymbol{\psi}_m(\tilde{\mathbf{y}}) dS_{\tilde{\mathbf{y}}} dS_{\tilde{\mathbf{x}}}. \quad (47)$$

In this sum all terms are regular except for $n=0$. This term $\tilde{\mathbf{U}}_{\Sigma\Sigma}^{oo}$ is nothing but the usual 3D-BEM term $\mathbf{U}_{\Sigma\Sigma}^o$ and can be computed using classical techniques (either singular integral procedures or regularisation techniques for homogeneous [6] or stratified domains [2–3]). The other terms are regular and can be computed using standard Gaussian formula. Equation (47) shows that they can be computed by simple translation of the source position $\tilde{\mathbf{y}}$ by $n\mathbf{l}$. The infinite sum is truncated when convergence is reached and criteria are given by Elhabre [19] and Elhabre *et al.* [20]. This convergence is much faster when damping is accounted for, thanks to the exponential decay of the Green's functions in this case.

4. Variability of the Incident Field

In this section, no fluctuation of the soil parameters is accounted for ($\Omega_\delta = \emptyset$) and only a finite number of bounded buildings with known properties is considered (Ω_b is bounded and deterministic). The incident field $\mathbf{u}_o^{\text{inc}}$ is random and modeled as a second order stochastic process. Since the solution $\mathbf{u}_b$ is a linear function of the incident field, linear filtering theory [28] characterizes directly $\mathbf{u}_b$ at any point in the structure as a function of $\mathbf{u}_o^{\text{inc}}$ on Σ . This is done using the transmission operator $\mathcal{T}_\Sigma$ given in equation (29).

As a non-homogeneous and non stationary stochastic process, the input field $\mathbf{u}_o^{\text{inc}}$ on Σ is usually expressed in the time domain as a function of an homogeneous stationary process $\mathbf{u}_{oo}^{\text{inc}}(\mathbf{x}, t)$ as follows:

$$\mathbf{u}_o^{\text{inc}}(\mathbf{x}, t) = g(t) \mathbf{u}_{oo}^{\text{inc}}, \quad (48)$$

where $g(t)$ is a deterministic modulation function. Moreover, since Σ is bounded, the spectral covariance of $\mathbf{u}_{oo}^{\text{inc}}$ denoted by $\mathbf{R}_{\text{inc}}(\mathbf{x}, \mathbf{x}', \omega)$ has the following representation:

$$\mathbf{R}_i(\mathbf{x}, \mathbf{x}', \omega) = \sum_{r=1}^{+\infty} (\sigma_r^{\text{inc}})^2 \mathbf{u}_{or}^{\text{inc}}(\mathbf{x}) \otimes \overline{\mathbf{u}_{or}^{\text{inc}}}(\mathbf{x}'), \quad \int_{\Sigma} \|\mathbf{u}_{or}^{\text{inc}}\|^2 dS = 1, \quad (49)$$

where $\mathbf{u}_{or}^{\text{inc}}$ are the eigenmodes of the integral operator of kernel $\mathbf{R}_{\text{inc}}$, $(\sigma_r^{\text{inc}})^2$ being the corresponding eigenvalues and $\overline{\mathbf{v}}$ denotes the conjugate of $\mathbf{u}$. In fact $\mathbf{u}_{or}^{\text{inc}}$ are modes of variation of the incident field, whereas σ_r^{inc} are the standard deviations on the amplitudes of these modes. The series (49) can be truncated above by a given number N since σ_r^{inc} is tending to zero for large r .

Then computing the deterministic solution $\mathbf{u}_{or}$ of the original problem for $\mathbf{u}_{or}^{\text{inc}}$ instead of $\mathbf{u}_o^{\text{inc}}$, the output becomes a linear expansion of these deterministic solutions:

$$\mathbf{u}_b = \sum_{r=1}^N \sigma_r^{\text{inc}} \theta_r \mathbf{u}_{or}, \quad \mathbf{u}_{or} = \mathcal{T}_\Sigma(\mathbf{u}_{or}^{\text{inc}}), \quad (50)$$

θ_r being uncorrelated random variables with unitary standard deviation. As a particular consequence, the evolutive standard deviation at a given point of the structure reads:

$$\sigma^2(t) = \int_{\mathbb{R}} \sum_{r=1}^N \sigma_r^2 \mathbf{u}_r(t, \omega) \cdot \mathbf{u}_r(t, \omega) d\omega, \quad (51)$$

$$\mathbf{u}_r(t, \omega) = \mathcal{F}(\mathbf{u}_{or} \delta_\omega * \hat{g}), \quad (52)$$

where $(\delta_\omega * \hat{g})(\omega') = \hat{g}(\omega' - \omega)$, $\hat{g}$ being the Fourier transform of g and $\mathcal{F}$ denoting the inverse Fourier transform with respect to ω' . The standard deviation $\sigma(t)$ provides the engineer with bounds on the amplitudes of the displacements of the structure.

5. Variability of Soil Parameters

In Section 4 it has been observed that a stochastic model for this incident field can be reduced to a set of deterministic incident fields. Therefore in this section $\mathbf{u}_o^{\text{inc}}$ is assumed to be a deterministic field. Moreover the fluctuations of the soil parameters are assumed to be restricted to a bounded region Ω_δ near the bounded structures Ω_b . These structures are assumed to be known. The perturbation of the soil parameters $\mathbf{p}_\delta$ is modeled as a centered second order Gaussian stochastic process on Ω_δ characterized by its covariance $\mathbf{R}_p$.

5.1. SPATIAL APPROXIMATION

Since Ω_δ is bounded, the random process $\mathbf{p}_\delta$ can be approximated by the following finite Karhunen-Loeve expansion [23]:

$$\mathbf{p}_\delta = \sum_{r=1}^N \theta_r \mathbf{p}_r, \quad (53)$$

where $\mathbf{p}_r$ are the first eigenmodes of the covariance operator associated to the kernel $\mathbf{R}_p(\mathbf{x}, \mathbf{x}')$ with eigenvalues equal to σ_r^2 :

$$\sigma_r^2 \mathbf{p}_r(\mathbf{x}') = \int_{\Omega_\delta} \mathbf{R}_p(\mathbf{x}, \mathbf{x}') \mathbf{p}_r(\mathbf{x}) dV(\mathbf{x}), \quad \int_{\Omega_\delta} \mathbf{p}_r^2 dV = \frac{1}{\sigma_r^2}. \quad (54)$$

In the Karhunen-Loeve expansion (53), θ_r are normalized centred uncorrelated random variables with values on the real axis $\mathbb{R}$. Operator $\mathcal{S}_\delta$ being a linear function of the physical parameters $\mathbf{p}_\delta$, the Lippman-Schwinger equation (20) reads:

$$\mathcal{U}_\delta \phi = \sum_{r=1}^N \theta_r \mathcal{S}_r(\mathcal{U}_\delta \phi - \mathbf{u}^{\text{inc}}), \quad (55)$$

where $\mathcal{S}_r$ is the deterministic scattering operator on Ω_δ having the properties $\mathbf{p}_r$.

This equation can be looked at in the context of wave propagation in a media with a random distribution of particles. In this case, the sum on r becomes an infinite sum on the particles, $\mathbf{p}_r(\mathbf{x})$ being the physical properties of particle number r and θ_r taking its values in $\{0, 1\}$ depending on the effective existence of the particle. Moreover cross-correlations between θ_r give statistical information on the particle distribution. Hence, equation (55) generalized the

classical approach used for a random particle distribution, to random fluctuation of physical parameters.

As far as spatial approximation is concerned, matrices $\mathbf{K}_\delta$ and $\mathbf{M}_\delta$ appearing in equation (35), are linear functions of the random variables θ_r :

$$\mathbf{K}_\delta = \sum_r \theta_r \mathbf{K}_r, \quad \mathbf{M}_\delta = \sum_r \theta_r \mathbf{M}_r. \quad (56)$$

$\mathbf{K}_r$ and $\mathbf{M}_r$ being deterministic matrices computed using the elastic parameters $\mathbf{p}_r$. Therefore, equation (34) now reads:

$$\mathbf{U} \mathbf{q}_\delta + \sum_r \theta_r \mathbf{S}_r (\mathbf{U}_\delta \mathbf{q}_\delta + \mathbf{u}^{\text{inc}}) = \mathbf{0}, \quad (57)$$

where $\mathbf{S}_r$ is a scattering matrix in the reference medium Ω_o due to the deterministic fluctuations of the elastic parameters $\mathbf{p}_r$ and is given by:

$$\mathbf{S}_r = \mathbf{U}_\delta^T (\mathbf{K}_r - \omega^2 \mathbf{M}_r). \quad (58)$$

5.2. APPROXIMATION OF THE RANDOM DIMENSION

Only approximation of the fields with respect to the space variables $\mathbf{x}$ have been considered so far. Using the Karhunen-Loeve expansion, it has been shown that $\mathbf{p}_\delta$ is a linear function of $\boldsymbol{\theta}$, the vector of the random variables θ_r . The vector $\mathbf{q}_\delta$ is a non linear function of $\boldsymbol{\theta}$ since it is a function of the inverse of a matrix depending on $\boldsymbol{\theta}$. The aim of the subsection is to deal with the approximation of this function with respect to $\boldsymbol{\theta}$, referred here as the random dimension of the problem.

5.2.1. Gaussian Hilbert spaces

From the mathematical point of view, the parameter space associated with $\boldsymbol{\theta}$ is assumed to be a finite dimension Gaussian Hilbert space for the classical weighted scalar product based on the Gaussian probability density functions $f_r(\theta_r)$ and it is denoted by H_p . The displacement field can be modeled as a stochastic process $\mathbf{u}(\mathbf{x}, \boldsymbol{\theta})$ belonging to the Gaussian Hilbert space $L^2(\Omega_p, V)$ for the scalar product $E\langle(\cdot, \cdot)\rangle$, $E\langle\cdot\rangle$ denoting the mathematical expectation [24, 27], V being the space of displacement fields on Ω with finite energy and $(\cdot, \cdot)$, the associated norm on this space.

5.2.2. The approximated stochastic problem

As far as discrete models are concerned, $\mathbf{q}_\delta$ belongs to $L^2(\Omega_p, \mathbb{R}^{3N_\delta})$ and this random vector can be approximated with bounded errors on a finite dimension basis $\{\varphi_k(\boldsymbol{\theta})\}_{k=1, n_k}$ with respect to $\boldsymbol{\theta}$ as follows:

$$\mathbf{q}_\delta(\boldsymbol{\theta}) = \sum_k^{N_k} \varphi_k(\boldsymbol{\theta}) \mathbf{q}_k. \quad (59)$$

Then multiplying equation (57) by any trial function $\varphi_{k'}(\boldsymbol{\theta})$ in the basis and taking the mathematical expectation leads to the following deterministic linear system:

$$\mathbf{U} \mathbf{q}_{k'} + \sum_{rk} \mathbf{U}_{rkk'} \mathbf{q}_k = \sum_n \mathbf{v}_{rk'}, \quad (60)$$

$$\mathbf{U}_{rkk'} = E\langle\theta_r\varphi_{k'}\varphi_k\rangle\mathbf{S}_r\mathbf{U}_\delta^T, \quad (61)$$

$$\mathbf{v}_{rk'} = E\langle\theta_r\varphi_{k'}\rangle\mathbf{S}_r\mathbf{u}^{\text{inc}}. \quad (62)$$

Basis functions $\varphi_k(\theta)$ can be either the Hermite polynomials for spectral methods or piecewise constant functions for Monte-Carlo simulations. In the former case the overall matrix is a sparse block matrix, whereas in the latter it is block diagonal. Indeed Monte-Carlo simulation simply consists in solving a large number of independent linear system for given values of the random variables θ_r .

5.2.3. Monte-Carlo simulations

Even though parallel computing may significantly reduce the computational effort, a large number of simulations have to be performed to span the parameter space. However since the proposed formulation consists in factorizing once the deterministic part of the problem using Integral Operators (wave propagation in the stratified half-space and the soil-structure interaction) the size of the final system is significantly reduced. Moreover the number of required simulations can be significantly reduced using variance reduction techniques and hypercube sampling [34].

Once $\mathbf{q}_k$, the realization of the random vector $\mathbf{q}_\delta$ is computed, the statistical properties of the stochastic process $\mathbf{U}_b(\mathbf{x}) = \{\mathbf{u}(\mathbf{x}, \boldsymbol{\theta}), \boldsymbol{\theta} \in \Omega_p\}$ can be estimated. Indeed from equation (38), $\mathbf{U}_b(\mathbf{x})$ is nothing but a linear function of $\mathbf{q}_\delta$. For example, given the second order statistics on $\mathbf{q}_\delta$, namely the mean value $\underline{\mathbf{q}}_\delta$ and the covariance matrix $\mathbf{R}_{\mathbf{q}_\delta}$:

$$\underline{\mathbf{q}}_\delta = \sum_k \mathbf{q}_k E\langle\varphi_k(\boldsymbol{\theta})\rangle, \quad \mathbf{R}_{\mathbf{q}_\delta} = \sum_k \mathbf{q}_k \otimes \overline{\mathbf{q}_k} E\langle\varphi_k^2(\boldsymbol{\theta})\rangle, \quad (63)$$

the second order statistics on $\mathbf{U}_b(\mathbf{x})$ are given by:

$$\underline{\mathbf{U}}_b(\mathbf{x}_b) = \mathbf{U}_\Sigma^b(\mathbf{x}_b)\mathbf{q}_o^{\text{inc}} + \mathbf{U}_\delta^b \underline{\mathbf{q}}_\delta, \quad (64)$$

$$\mathbf{R}_{\mathbf{U}_b}(\mathbf{x}_b, \mathbf{x}'_b) = \mathbf{U}_\delta^b(\mathbf{x}_b)\mathbf{R}_{\mathbf{q}_\delta}\overline{\mathbf{U}_\delta^b}^T(\mathbf{x}'_b). \quad (65)$$

Thus using this approach the computation of realistic models such as a reactor building resting on a stratified half-space perturbed by large random fluctuation under the foundation can be performed in a few hours on a desktop computer as proposed in the next subsection.

5.3. APPLICATION

The structure studied is a nuclear reactor building resting on a circular spread footing of radius $R \simeq 25$ m (see Figure 2). The analysis is restricted to the case of random Lamé's moduli for the soil. This means that the elasticity tensor in the soil medium is isotropic and depends only on two parameters λ and μ written in the perturbation zone Ω_δ as:

$$\lambda(\mathbf{x}) = \lambda_o(z) + \lambda_\delta(\mathbf{x}), \quad \mu(\mathbf{x}) = \mu_o(z) + \mu_\delta(\mathbf{x}),$$

where λ_o and μ_o are Lamé's moduli for the soil mean stratification, function of the depth z . The structural displacements on a fixed basis are expanded along the first 30 eigenmodes $\mathbf{v}_n$, with constant critical damping rates $\zeta_A = 7\%$. The stratified soil medium (without heterogeneity) consists of one layer of thickness 40 m and a bedrock. Mechanical characteristics are respectively: $\lambda_o = 5.74 \times 10^3$ and 1.62×10^4 MPa, $\mu_o = 1.81 \times 10^3$ and 110^4 MPa,

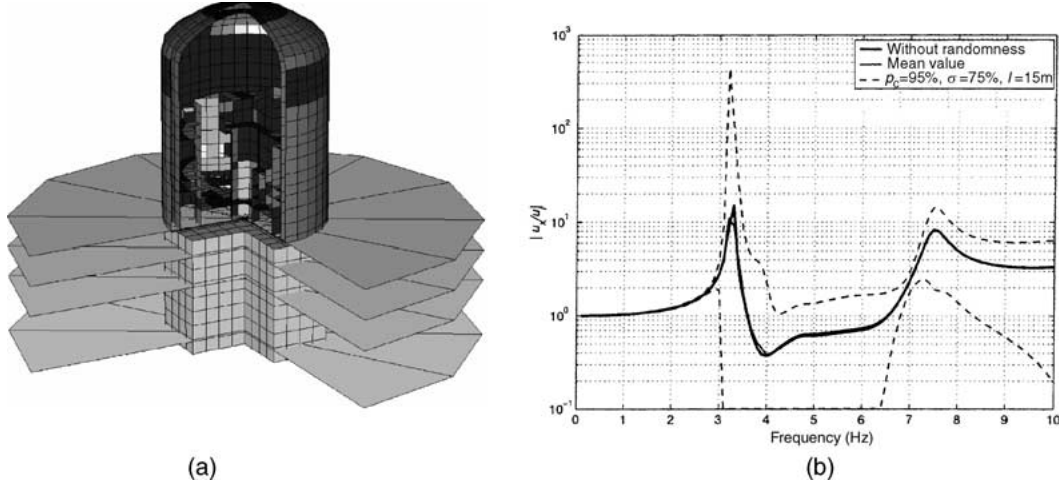


Figure 2. Influence of the soil variability on the seismic response of a reactor building: (a) the mesh, (b) the frequency response on top for a standard deviation of 75 and 50% and a correlation length of 15 m. Mean values and limits for 95% confidence are plotted. Three thousand trials were used and 40 modes for the Karhunen-Loeve expansion of the perturbations of Lamé's moduli.

$\rho_o = 2400 \text{ kg m}^3$. The random soil heterogeneity occupies a cubic block $40 \times 40 \times 40 \text{ m}^3$ and its auto-correlation function is given by:

$$R(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|}{\ell}\right), \quad (66)$$

where σ is the standard deviation and ℓ the correlation length.

Figure 2 displays the log-scaled amplitude of the horizontal displacement on top of the structure normalized by the incident wave amplitude. Mean values and confidence intervals are plotted, the latter being defined by the upper and lower envelopes such that:

$$P(|u_x - \underline{u}_x| < U_c) \geq p_c,$$

with a confidence level p_c and $\underline{u}_x = E < u_x >$ the mean value. The upper and lower envelopes are constructed using Chebychev's inequality which results in U_c such that:

$$p_c = 1 - \frac{E < |u_x - \underline{u}_x|^2 >}{U_c^2}.$$

As shown on Figure 2, the dispersion of the structural response for a vertical incident S-wave is large but still some conclusions can be drawn. In particular, a reduction of the mean structural vibration amplitude up to 15% is achieved with a confidence of 95%. The fundamental eigenfrequency of the coupled system is almost unchanged at low frequencies whereas slight discrepancies can be observed for the higher coupled mode around 7.5 Hz.

6. Variability of Unbounded Structures

This section focuses on the randomness of the distribution of buildings and thus the soil and the incident field are assumed to be deterministic. When Ω_b is bounded, the method

proposed in Section 5 to account for the soil variability applies with very little modification. In particular, as far as the numerical approximation of the Ω_b domain is concerned, the standard Stochastic Finite Elements [23] can be used. The case of no correlation between a finite number of buildings is first addressed and then extended to an infinite homogeneous random distribution. Finally, the scattering of a seismic incident field inside a ‘random city’ is analysed.

6.1. SMALL CORRELATION LENGTH

It is first assumed that Σ is decomposed into N_r foundations on which N_t different types of buildings can rest. Assuming no correlation between the buildings, the K-L expansion of the second order stochastic field $\mathbf{p}_b$ reads:

$$\mathbf{p}_b(\mathbf{x}) = \sum_{r=1}^{N_r} \left\{ \sum_{t=1}^{N_t} \mathbf{p}_t(\mathbf{x} - \mathbf{x}^r) \theta_t^r \right\} \delta^r(\mathbf{x}_\Sigma), \quad (67)$$

δ^r being the characteristic function of foundation number r centred at $\mathbf{x}^r$ and $\mathbf{x}_\Sigma$ being the projection of $\mathbf{x}$ on Σ . The deterministic fields $\mathbf{p}_t(\mathbf{x})$ give the physical properties of a finite set reference buildings indexed by t . θ_t^r are the components of random vectors $\boldsymbol{\theta}^r$ of size N_t . These components take their values in $\{0, 1\}$ and satisfy $\sum_{t=1, N_t} \theta_t^r = 1$. In fact, the N_r random vectors $\boldsymbol{\theta}^r$ are the images of the same random vector θ_o associated to N_t possible events (N_t types of buildings) and characterized by the probability coefficients f_t for all t .

6.2. HOMOGENEOUS DISTRIBUTION AND PERIODIC MODELS

When extended to an infinite number of buildings, $N_r \rightarrow +\infty$, $\mathbf{p}_b(\mathbf{x})$ is clearly homogeneous with respect to $\mathbf{x}_\Sigma$ and the spatial correlation is limited to the size of one foundation. The sum on r can be decomposed into two scales, a large scale consisting of periodic cells indexed by $\mathbf{n}$, and a short scale inside each cell, where buildings are indexed by $\tilde{r}$ ranging from 1 to $\tilde{N}_r$:

$$\mathbf{p}_b(\mathbf{x}) = \sum_{\mathbf{n} \in \mathbb{Z}^2} \sum_{\tilde{r}=1}^{\tilde{N}_r} \sum_{t=1}^{N_t} \mathbf{p}_t(\tilde{\mathbf{x}} - \tilde{\mathbf{x}}^r) \theta_t^n \delta^{\tilde{r}}(\tilde{\mathbf{x}}_\Sigma), \quad (68)$$

where r is a known function of $\tilde{r}$ and $\mathbf{n}$ and where $\tilde{\mathbf{x}} = \mathbf{x} + \mathbf{n}l$. The Floquet transform of $\mathbf{p}_b$ is then given by:

$$\mathbf{p}_b(\tilde{\mathbf{x}}, \boldsymbol{\kappa}) = \sum_{\tilde{r}=1}^{\tilde{N}_r} \sum_{t=1}^{N_t} \mathbf{p}_t(\tilde{\mathbf{x}} - \tilde{\mathbf{x}}^r) \left\{ \sum_{\mathbf{n} \in \mathbb{Z}^2} \theta_t^n e^{i\boldsymbol{\kappa} \cdot \mathbf{n}l} \right\} \delta^{\tilde{r}}(\tilde{\mathbf{x}}_\Sigma). \quad (69)$$

The mean value of $e^{i\boldsymbol{\kappa} \cdot \mathbf{n}l}$ with respect to $\mathbf{n}$ is almost surely equal to 0 as long as the period l is large compared to the size of one building and $\boldsymbol{\kappa} \neq \mathbf{0}$. As a consequence, $\mathbf{p}_b(\mathbf{x})$ is almost surely periodic and on the reference cell it can be equal to any random distribution of buildings with probability coefficients f_t and thus:

$$\mathbf{p}_b(\tilde{\mathbf{x}}, \boldsymbol{\kappa}) \approx \delta(\boldsymbol{\kappa}) \sum_{\tilde{r}=1}^{\tilde{N}_r} \sum_{t=1}^{N_t} \mathbf{p}_t(\tilde{\mathbf{x}} - \tilde{\mathbf{x}}^r) \sum_{\mathbf{n} \in \mathbb{Z}^2} \theta_t^n \delta^{\tilde{r}}(\tilde{\mathbf{x}}_\Sigma). \quad (70)$$

This result holds in the general case as long as the correlation length of $\mathbf{p}_b$ remains small with respect to the size of the reference cell, namely the periods l_1 and l_2 .

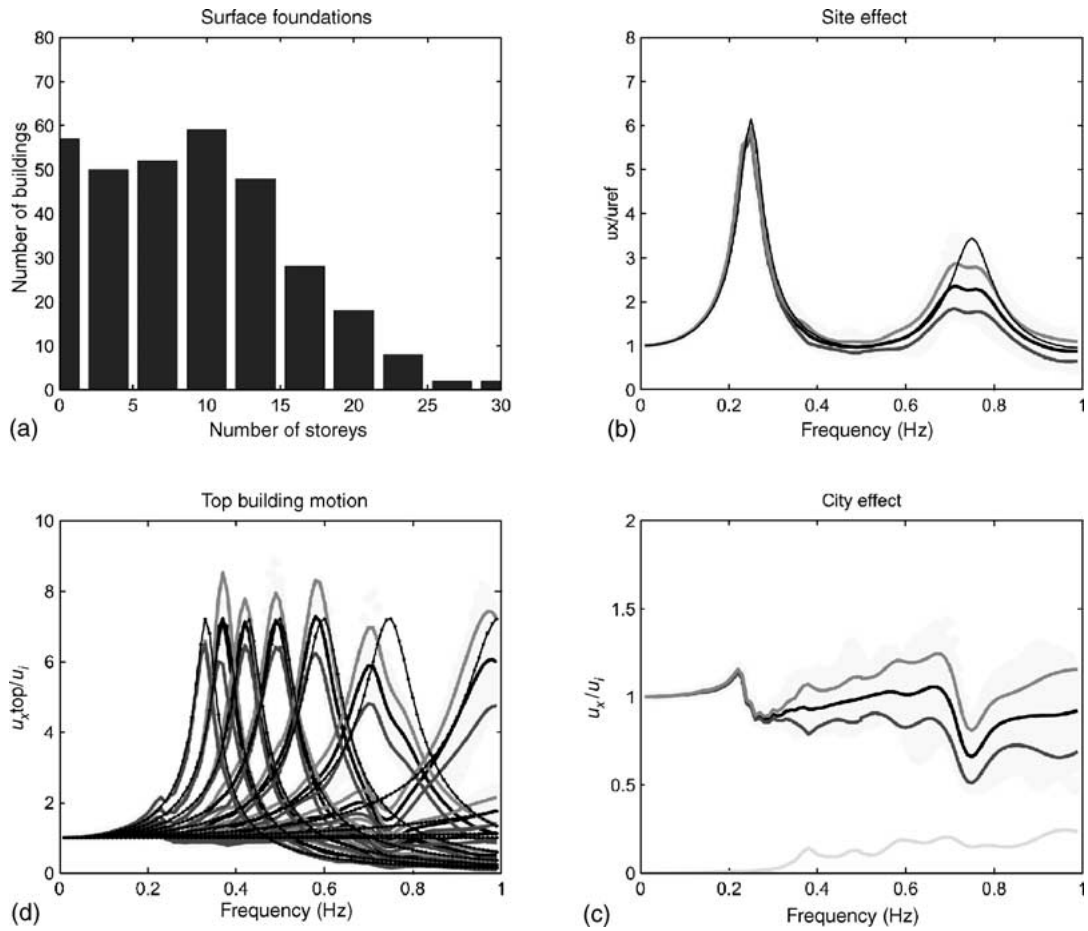


Figure 3. The seismic response of 324 buildings distributed randomly on top of a layered half-space (individual results produce the gray areas, thick lines indicate the mean values, the mean values $\pm$ the standard deviation): (a) the building repartition, (b) spectral ratios between the response on top of the layer and on top of a free bedrock (the thin line indicates the response without the buildings), (c) spectral ratios between the response on top of the layer and the free-field at the same place and the standard deviation of the scattered field, (d) spectral ratios between the top building response and the free field (the thin lines with dots indicate the top to bottom spectral ratio of the buildings).

Since $\mathbf{p}_b(\mathbf{x})$ is almost surely periodic, the SSI interaction problem can be solved using the periodic approach described in Section 3 for different possible distributions of buildings on the reference cell. Statistics on the displacement fields are then obtained using formula (63). Due to the sum over $\mathbf{n}$ in formula (69) high convergence rates of the Monte-Carlo simulation are expected. Moreover, as long as the cell is large enough to incorporate a significant number of buildings of each type, statistics inside the cell itself give valuable estimates of the first statistical moments of the displacement fields.

6.3. THE CITY-SITE EFFECT

The proposed method has been used to study the effect of a random distribution of buildings on the seismic response of a given site. For this application the random building distribution is generated by an heuristic method (pieces of storey are randomly distributed inside

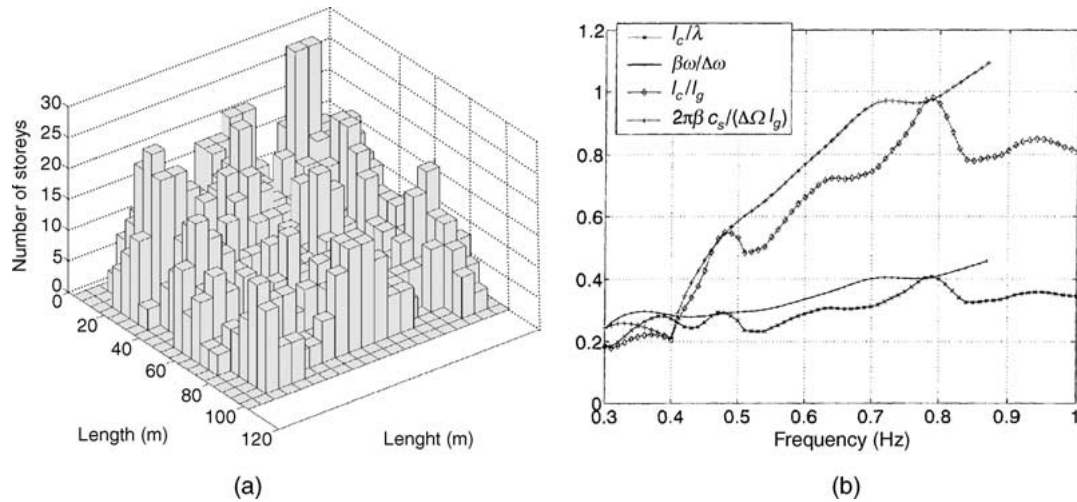


Figure 4. Correlation on top of the half-space of the field scattered by a random city (a). (b) Correlation length l_c normalized either by the wavelength λ or by the correlation length of the building distribution l_g and correlation frequency $\Delta\omega$ normalized either by $\beta\omega$ or by $\beta\omega\lambda/l_g$, with β the damping ratio of the buildings.

the cell) satisfying in a mean sense architectural requirements (the process is stopped when an average built surface is reached on the cell) and structural engineering constraints (The fundamental natural period of a building is a linear function of the number of storeys). Once the numerical solution is computed, statistical analysis is performed on the seismic field showing both diffusion and localisation of the seismic field as shown in Figures 3 and 4 [14]. More precisely these results show that the influence of interactions between buildings is increasing with the frequency. These interactions slightly reduce the mean response of the site but do not significantly modify the response of the buildings. Moreover, the correlation length of the scattered field is inversely proportional to the frequency and is almost equal to one quarter of a wavelength. This result is in accordance with localisation phenomena occurring in a random media [32]. The coherency on the frequency response has also been computed showing increases in the duration of the same order as the characteristic relaxation time of the buildings [13, 16].

7. Conclusions

Three numerical methods aiming at accounting for randomness of either the incident field or the physical parameters have been proposed. These methods are based on a detailed mathematical and numerical modelling of the deterministic problem using both integral operators and variational approaches. In particular explicit expressions of the transmission and the scattering operator have been worked out for the first time in the context of soil–structure interaction.

The classical Lippman-Schwinger equation used for random particle distributions has been generalized to general random fluctuations of the soil parameters. A numerical version of this equation, first proposed in Savin [34] and Savin and Clouteau [35], has been more fully developed. Both the spatial and the random dimensions are discretized and the scattering operator on the fluctuation has been introduced. Moreover a non-singular integral equation accounting for heterogeneous media have has proposed.

As far as unbounded random fluctuations of the physical parameters are concerned, a deterministic method to account for bidimensional network has been proposed for the first time in a SSI context. The usefulness of such models to account for unbounded homogeneous fluctuations has been shown in a second step.

Although restricted for the sake of clarity to surface foundations, the proposed methods apply for embedded ones including other types of structures such as tunnels, dams, etc. Moreover, even though presented in sequence, randomness of the incident field, the soil parameters or the distribution of buildings can be combined. In closing, the scope of applicability of the proposed method is much wider than seismic soil–structure interaction.

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